Chaitanya Bharathi Institute of Technology (CBIT)

Department of Computer Science and Engineering

FUNDAMENTALS OF DATA SCIENCE

Comprehensive Case Study Report (CS-2 & CS-5)

Part I: Customer Segmentation using Clustering (East-West Airlines)
Part II: Wine Quality Prediction via Multiple Linear Regression

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PART I: Case Study 2 - Customer Segmentation using Clustering

1. Problem Statement and Objectives

East-West Airlines collected operational flight, mileage accrual, credit card partnership transactions, and enrollment tenure data for 3,999 frequent flyer accounts. The airline seeks to discover natural customer segments to tailor marketing campaigns, tier incentives, and loyalty redemption options rather than treating all members uniformly.

2. Dataset Preprocessing & Outlier Handling

The dataset contains 3,999 records across 12 features with 0 missing values. Feature ID# was dropped as an arbitrary identifier. The target loyalty indicator Award? was withheld from clustering and reserved strictly for post-clustering profiling. Continuous mileage features exhibited severe right-skewness (cc3_miles: +17.20, Balance: +5.00). Applying np.log1p normalized distributions. Outliers were removed using $|z| > 3$ on log-transformed variables, removing 376 extreme records (9.40%) and retaining 3,623 clean records without dropping legitimate high-tier flyers. Features were subsequently standardized using StandardScaler.

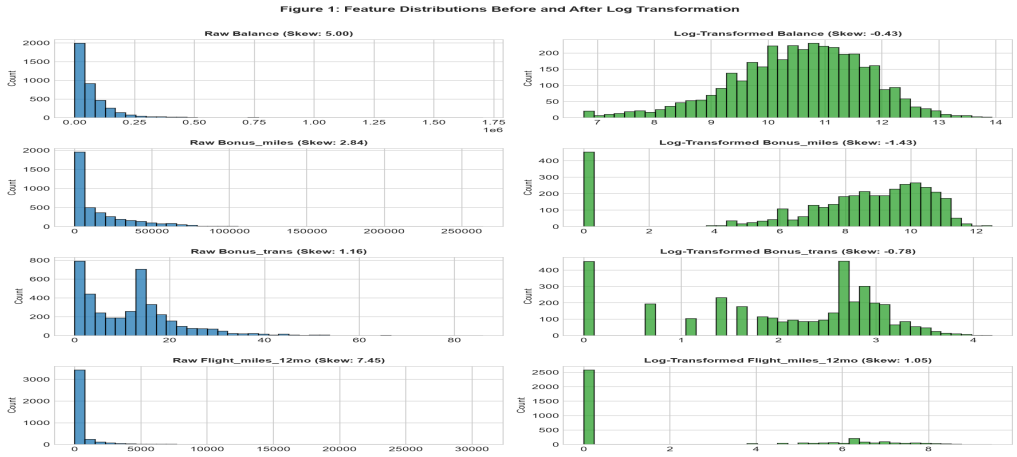


Figure 1: Distribution of key mileage features before and after log1p transformation

3. Exploratory Data Analysis & Structure

Correlation analysis revealed that credit card mileage (cc1_miles) correlates strongly with bonus miles (r = 0.83) and bonus transactions (r = 0.60), distinguishing credit card spenders from frequent flyers. Initial 2D PCA showed a dense continuous central customer core with extended arms representing elite flyers.

4. Clustering Algorithms and Model Evaluation

Three distinct clustering algorithms were systematically evaluated on the standardized feature matrix:
1. **K-Means:** Evaluated across K = 1 to 10. Both the inertia elbow curve and silhouette score profile indicated an optimal partition at K = 4 (Inertia = 12,568; Silhouette = 0.3199).
2. **Hierarchical (Agglomerative):** Evaluated across Ward, Complete, Average, and Single linkage. Ward linkage produced the most cohesive tree with the highest silhouette score (0.3317 at K = 4), while Single linkage suffered from chaining (silhouette = 0.2641).
3. **DBSCAN:** Evaluated using 5-nearest neighbor knee analysis. An eps of 0.879 with min_samples = 10 separated 3 dense customer cores while setting aside 220 noise points (6.07%).

Figure 4: K-Means Optimal K Selection via Elbow and Silhouette Analysis

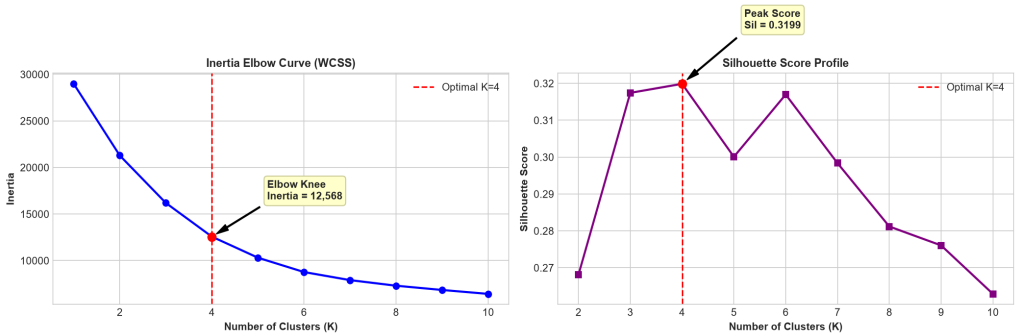


Figure 2: K-Means optimal K selection via inertia elbow and silhouette score analysis

Table 1: Clustering Algorithm Evaluation Comparison

Algorithm	Clusters	Silhouette	Davies-Bouldin	Calinski-Harabasz
K-Means	4	0.3199	0.9021	1575.63
Hierarchical (Ward)	4	0.3317	0.8735	1410.12
DBSCAN	3	0.3353	1.1554	1512.32

Figure 7: 2D PCA Visualizations of Customer Partitions Across All 3 Algorithms

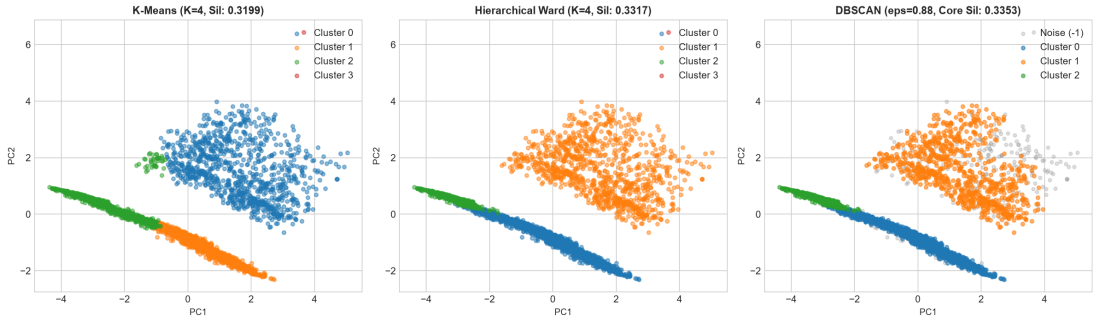


Figure 3: 2D PCA cluster visualizations comparing K-Means, Hierarchical (Ward), and DBSCAN

5. Customer Segment Personas & Marketing Recommendations

K-Means clustering at K = 4 revealed four distinct customer personas based on unscaled feature averages and award redemption rates:

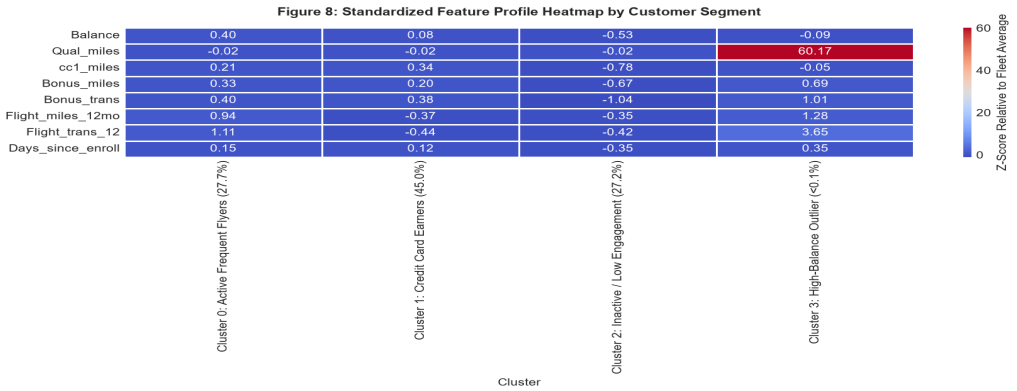


Figure 4: Standardized feature profile heatmap across K-Means customer segments

Customer Personas Identified:

1. **Cluster 0: Active Frequent Flyers (27.7% of accounts):** Characterized by high flight activity (1,076 flight miles/yr, 3.2 flight transactions) and a high Award flight redemption rate of 54.2%. *Strategy:* Retain tier status with complimentary lounge access, priority check-in, and seat upgrade certificates.
2. **Cluster 1: Credit Card Loyalty Earners (45.0% of accounts):** Characterized by high credit card spend (20,900 bonus miles, 15.5 transactions) but near-zero actual flight activity. *Strategy:* Offer targeted award seat promotions (e.g. 20% mileage discount on flights) to convert card spenders into active airline travelers.
3. **Cluster 2: Inactive / Low Engagement Members (27.2% of accounts):** Characterized by low balances (22,418 miles), minimal transactions, and low award rate (11.8%). *Strategy:* Re-engagement campaigns, entry-level mileage incentives, and account expiration reminders.
4. **Cluster 3: Elite Outlier Accounts (<0.1%):** High-volume qualifying accounts requiring dedicated corporate concierge support.

Practical Limitations of Clustering:

1. Cluster count K = 4 involves mathematical trade-offs between inertia reduction and segment interpretability.
2. DBSCAN struggled with non-uniform density, grouping 94% of customers into a single broad cluster while labeling 6% as noise.
3. PCA 2D projections retain ~40% of variance, meaning 60% of high-dimensional geometric separation is compressed.

PART II: Case Study 5 - Wine Quality Prediction via Multiple Regression

1. Problem Statement and Objectives

Wine quality assessment traditionally relies on expert sensory tasting panels, which is expensive, slow, and prone to taster palate fatigue. The objective of this case study is to determine whether laboratory-measured physicochemical attributes can predict wine quality ratings using statistical hypothesis testing and Multiple Linear Regression.

2. Dataset Description & Hygiene

The red wine variant of the Portuguese Vinho Verde dataset (Cortez et al., 2009) contains 1,599 observations across 11 physicochemical inputs and 1 quality target. Data hygiene checks identified 0 missing values and 240 duplicate rows (15.01%). Duplicate rows were removed prior to model training to prevent data leakage across train/test splits, yielding 1,359 clean, unique wine samples.

Table 2: Descriptive Statistics and Skewness (Task 1a)

Attribute	Mean	Median	Mode	Std Dev	Skewness
alcohol	10.4323	10.20	9.50	1.0821	0.8598
volatile acidity	0.5295	0.52	0.50	0.1830	0.7293
residual sugar	2.5234	2.20	2.00	1.3523	4.5482

Task 1b: Relative Dispersion via Coefficient of Variation (CV)

Residual Sugar exhibited the highest relative variability (CV = 53.59%), compared to Volatile Acidity (34.57%) and Alcohol (10.37%). CV is the mathematically appropriate comparative measure because standard deviation depends on measurement scale (% vol vs g/dm3), whereas CV normalizes dispersion by dividing standard deviation by the mean.

Task 1c: Pearson Correlation Hypothesis Test (Alcohol vs Quality)

A formal two-tailed test was performed (H0: r = 0 vs H1: r != 0, alpha = 0.05). The Pearson correlation was r = +0.4803 with p-value = 2.28e-79. Since p < 0.05, we reject H0 and conclude that a statistically significant, moderate positive linear relationship exists.

3. Exploratory Data Analysis & Feature Relationships

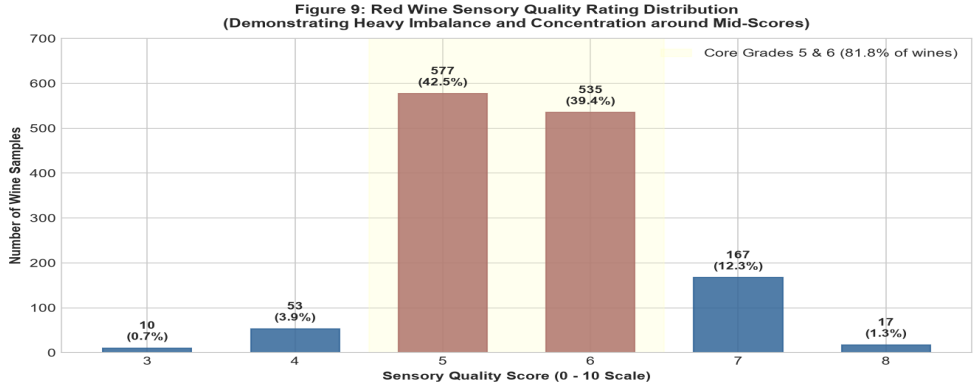


Figure 5: Wine sensory quality distribution showing heavy imbalance around scores 5 and 6 (81.8% of wines)

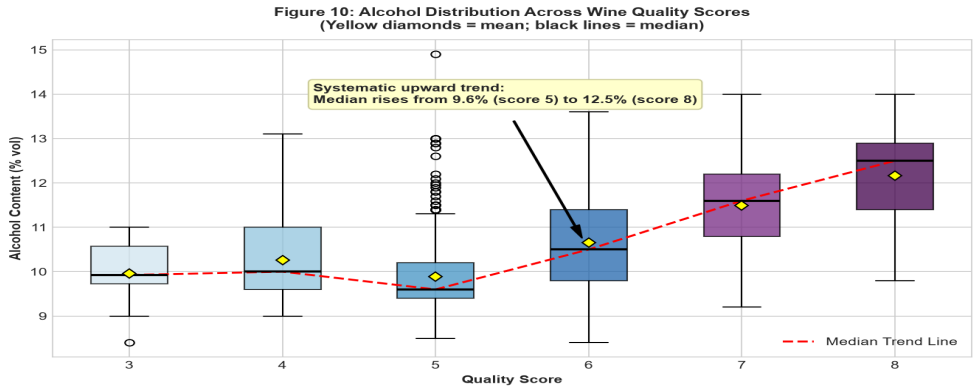


Figure 6: Alcohol distribution across quality tiers showing median upward trend from 9.6% to 12.5%

Task 2d: Correlation Ranking with Quality

Ranking all 11 attributes by absolute Pearson correlation identified the top two predictors: (1) **Alcohol (r = +0.4803)**: Positive association reflecting ripe fruit and proper fermentation; (2) **Volatile Acidity (r = -0.3952)**: Negative association reflecting unpleasant vinegar spoilage caused by acetic acid bacteria. Noticeable collinear pairs include fixed acidity with density (r = +0.67) and fixed acidity with pH (r = -0.68).

4. Multiple Linear Regression & Model Diagnostics

A multiple linear regression model was fitted on an 80:20 train-test split (1,087 train, 272 test, random_state=42). StandardScaler was fitted strictly on the training set to prevent data leakage.

Table 3: Multiple Linear Regression Test-Set Performance Metrics

Metric	Value	Interpretation
MAE	0.5041	Average absolute error in quality points
MSE	0.431	Average squared error (penalises large errors more)
RMSE	0.6565	Root MSE in quality points (same unit as target)
R2 (test)	0.3915	Fraction of quality variance explained by the model
R2 (train)	0.352	Train R2 for overfitting check

Multicollinearity Diagnosis (VIF Analysis):

Variance Inflation Factor (VIF) evaluation revealed moderate-to-high multicollinearity for Fixed Acidity (VIF = 7.95) and Density (VIF = 6.31). While predictions remain unbiased, individual coefficients for these collinear features must be interpreted with caution. Statsmodels OLS confirmed that volatile acidity, chlorides, total sulfur dioxide, sulphates, and alcohol are statistically significant ($p < 0.05$).

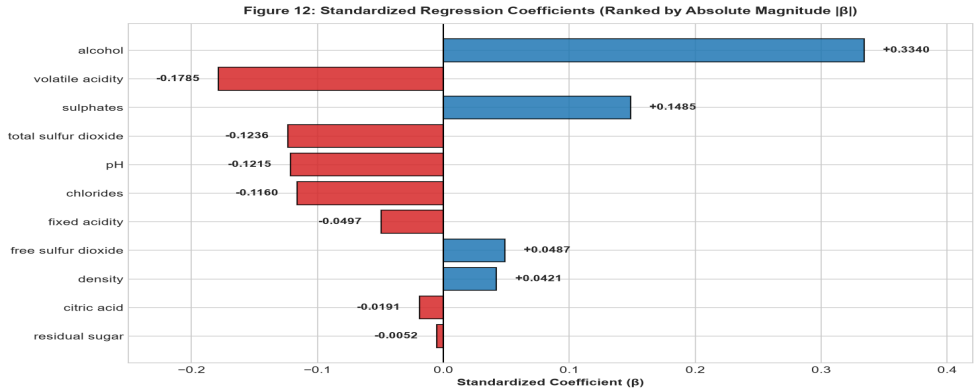


Figure 7: Standardized regression coefficients ranked by absolute impact |β| with direction coding

5. Model Diagnostics & Business Decision-Making

Figure 13: Model Predictive Diagnostics and Residual Structure

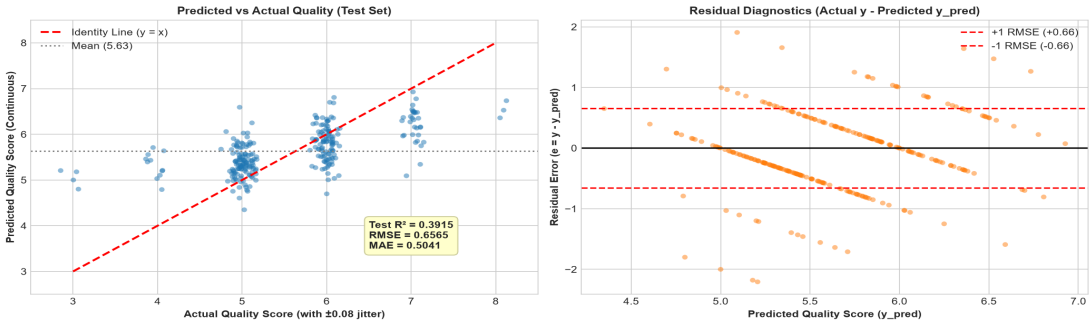


Figure 8: Predicted vs actual quality (left) and residual plot showing discrete score tier stripes (right)

Diagnostic Takeaways:

Continuous linear regression compresses predictions into a narrow band between 5.0 and 6.5, shrinking extreme scores toward the training mean (5.63). The residual plot exhibits distinct downward-sloping parallel stripes (e = k - y_pred) corresponding to discrete integer ratings.

Task 4: Business Decisions & Recommendations:

1. **Automated Lab Screening Measurements:** The winery should implement rapid automated testing on (a) **Alcohol Content** (strongest positive impact, beta = +0.3340, $p < 0.0001$) and (b) **Volatile Acidity** (strongest negative spoilage impact, beta = -0.1785, $p < 0.0001$).
2. **Can the Model Replace Sensory Panels? No.** With R2 = 0.3915 (~39% variance explained) and RMSE = 0.6565 quality points, prediction errors frequently cross an entire quality rating tier. The model should serve strictly as a pre-screening filter to catch flawed vats before submitting samples to human tasting panels.
3. **Real-World Limitations:** Restriction to red Vinho Verde wines, ordinal nature of subjective sensory ratings, and severe class imbalance (81.8% in scores 5-6).

6. Conclusion & References

This unified practical investigation applied unsupervised and supervised data science methodologies to real-world datasets. In Case Study 2, K-Means clustering (K = 4) and Hierarchical Ward linkage successfully segmented airline customers into actionable operational tiers. In Case Study 5, multiple linear regression achieved stable out-of-sample prediction (RMSE = 0.66, R2 = 0.39), identifying alcohol and volatile acidity as the primary chemical determinants of wine quality.

References:

1. Shmueli, G., and Bruce, P. C. (2016). *Data Mining for Business Analytics: Concepts, Techniques, and Applications in Python*. John Wiley & Sons.
2. Cortez, P., Cerdeira, A., Almeida, F., Matos, T., and Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties. *Decision Support Systems*, Elsevier, 47(4):547-553.
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4. Seabold, S., and Perktold, J. (2010). Statsmodels: Econometric and statistical modeling with Python. *Proceedings of the 9th Python in Science Conference*.
5. Project Repository: